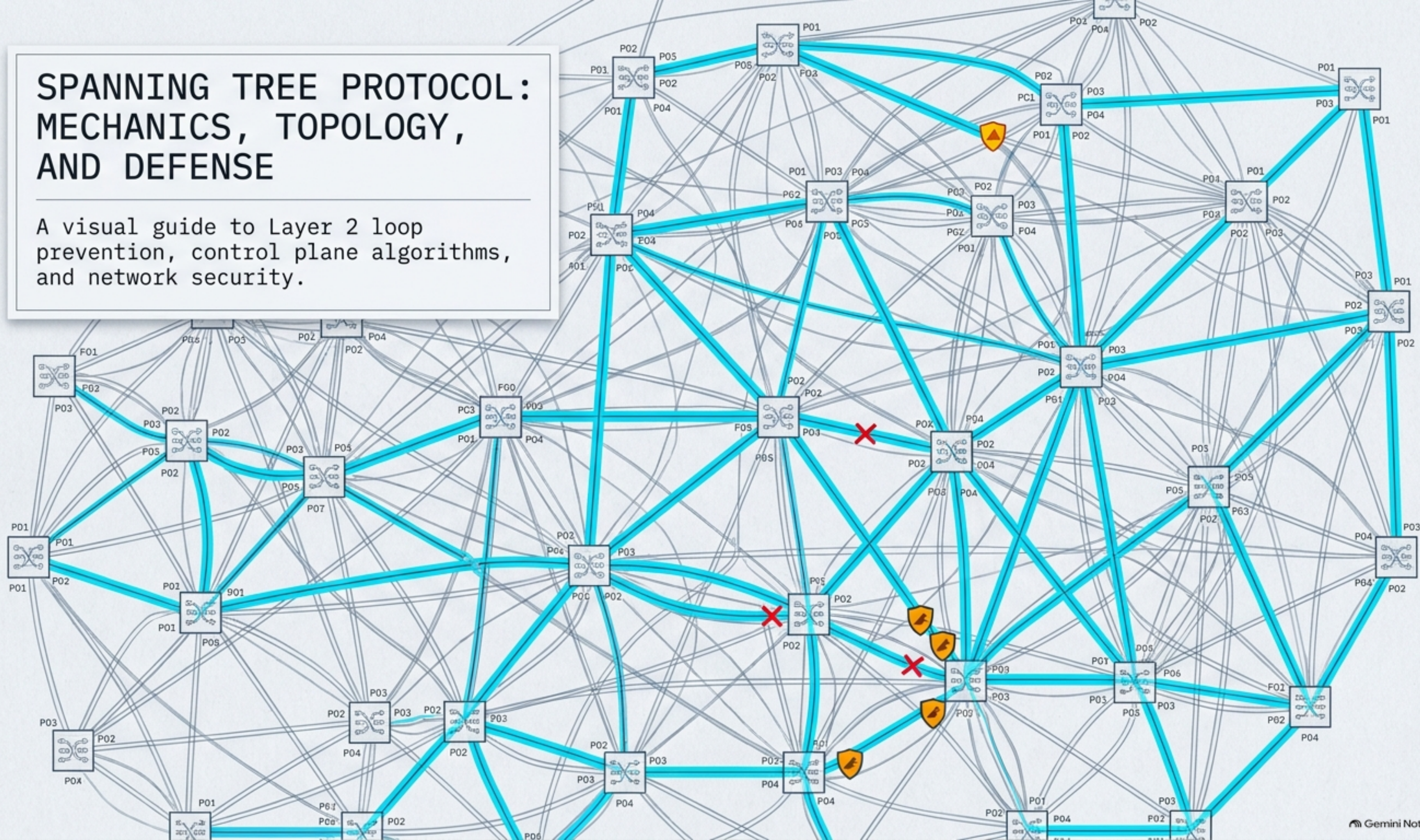
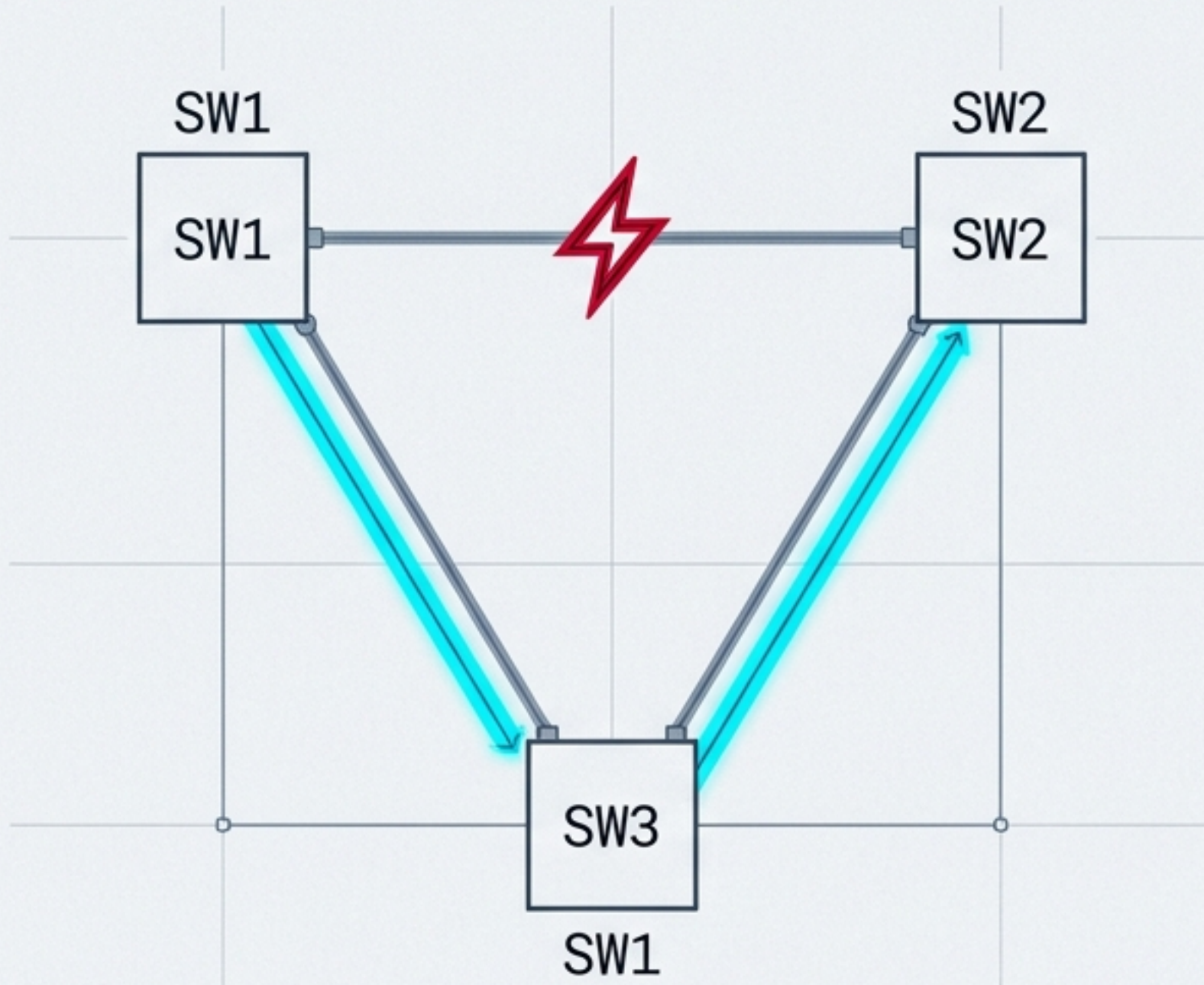


SPANNING TREE PROTOCOL: MECHANICS, TOPOLOGY, AND DEFENSE

A visual guide to Layer 2 loop prevention, control plane algorithms, and network security.

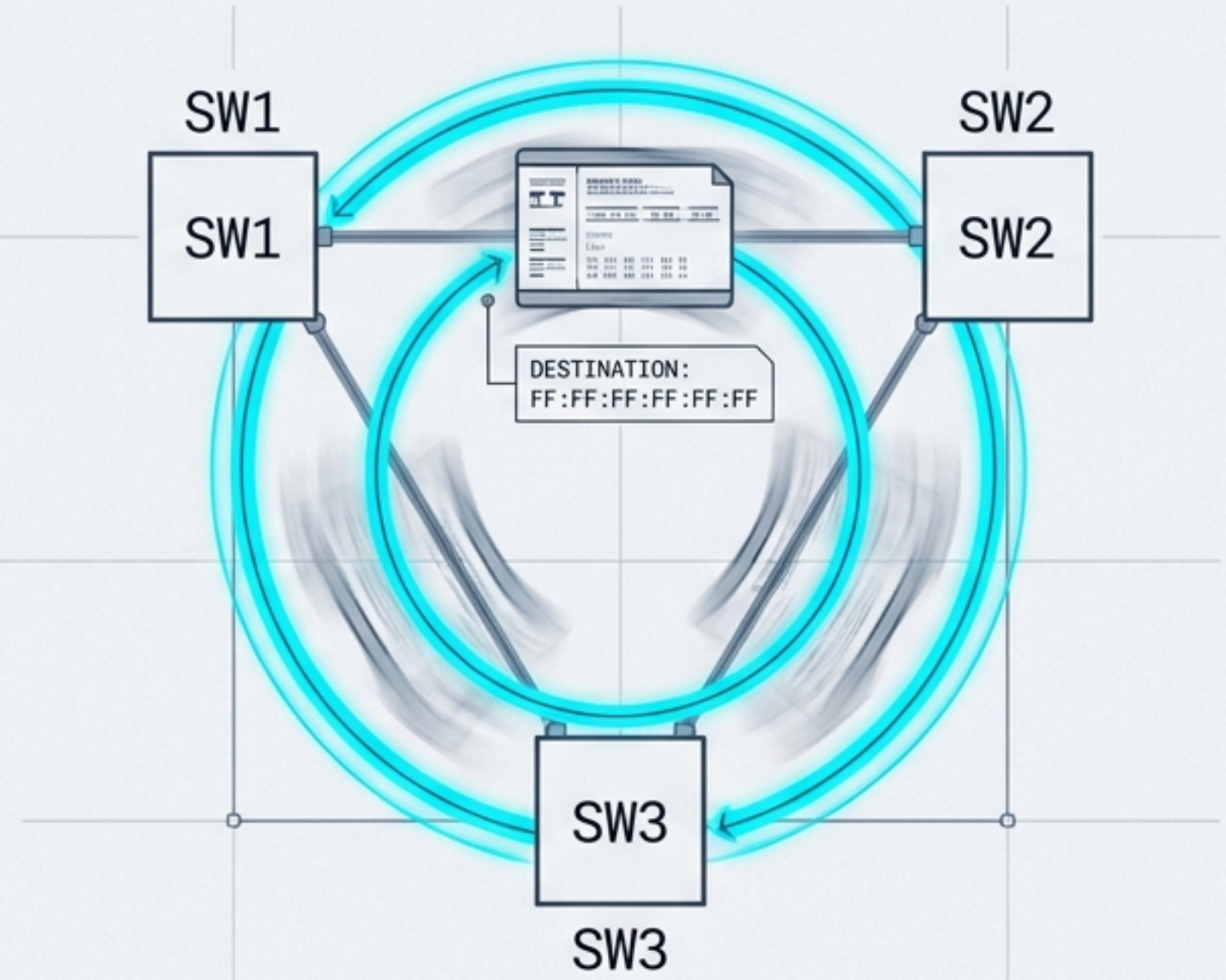


THE GOAL: REDUNDANCY & RESILIENCE



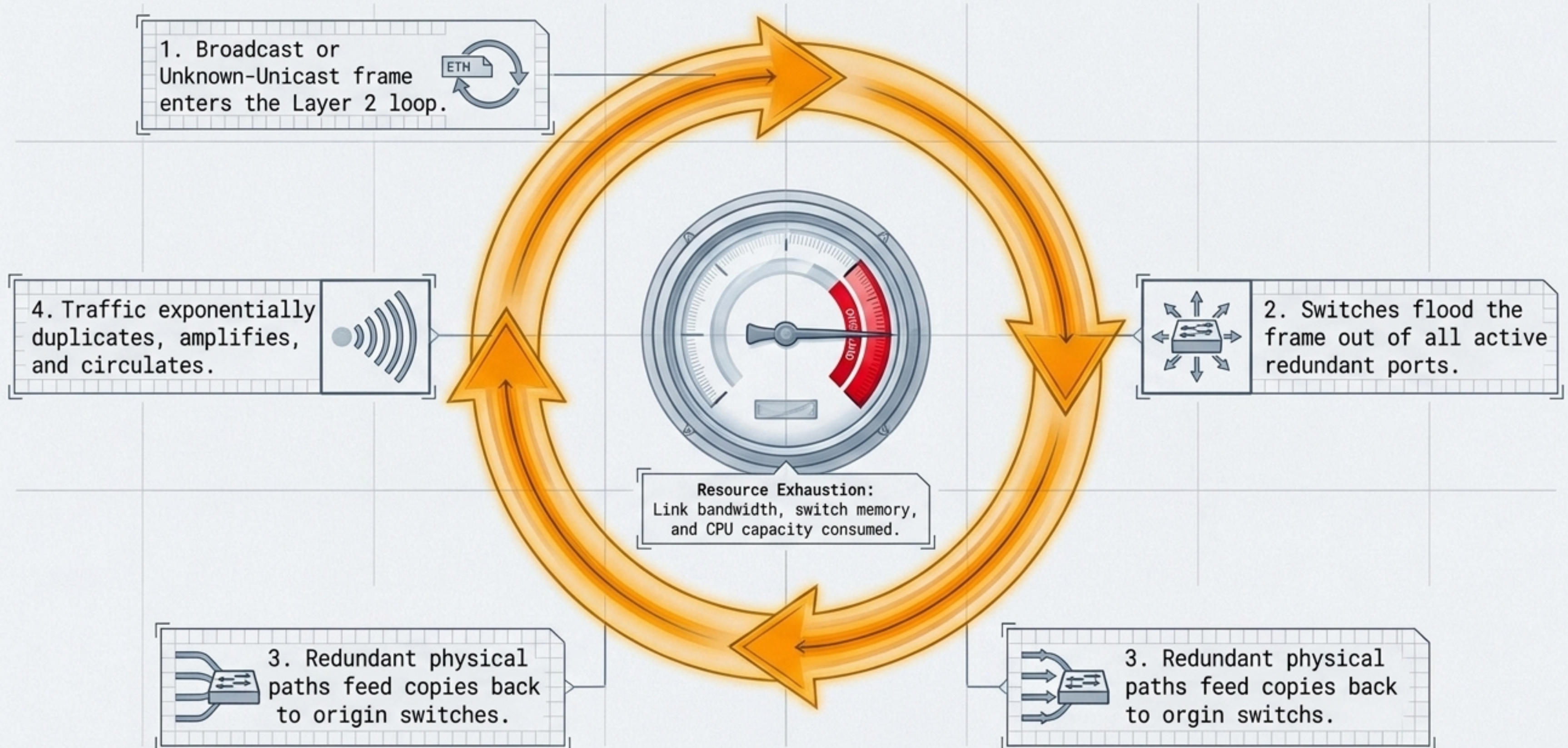
Primary path fails →
→ Alternate path remains →
→ Connectivity continues.

THE FLAW: INFINITE FORWARDING LOOPS

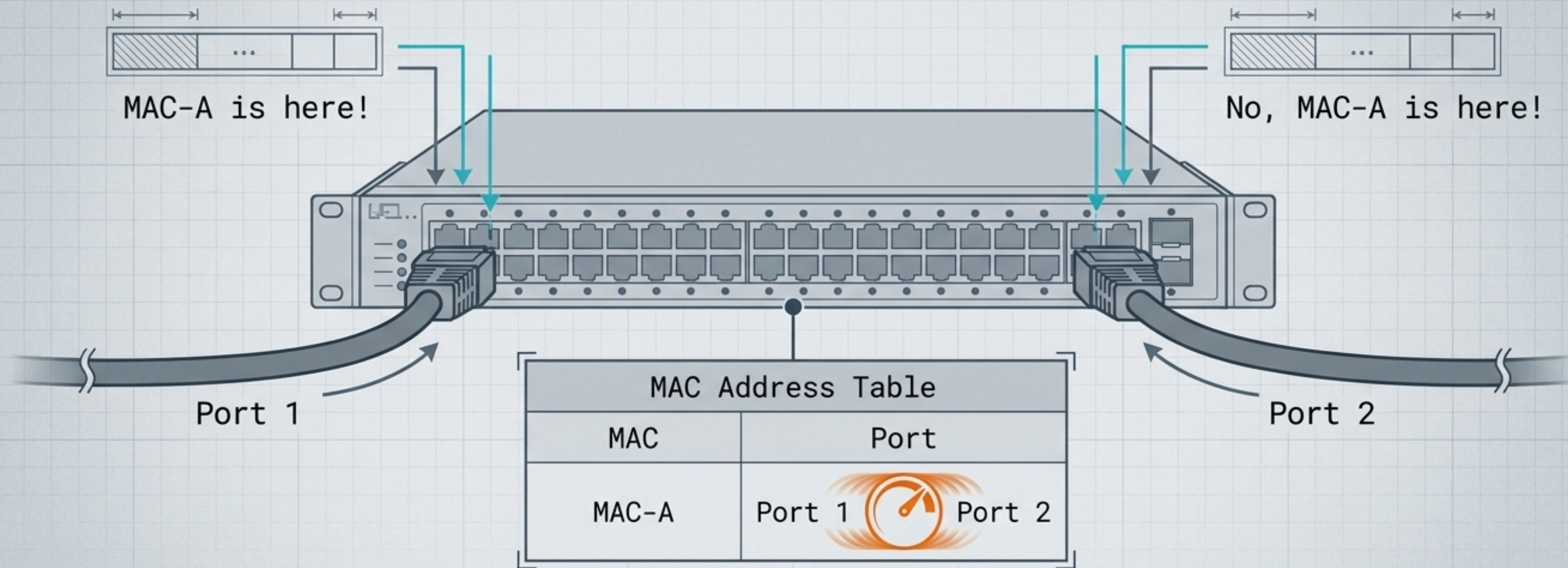


Unlike IP packets, Ethernet frames lack a Time-To-Live (TTL) mechanism. Unrestricted flooding creates infinite circulation.

The compounding failure of a Layer 2 broadcast storm

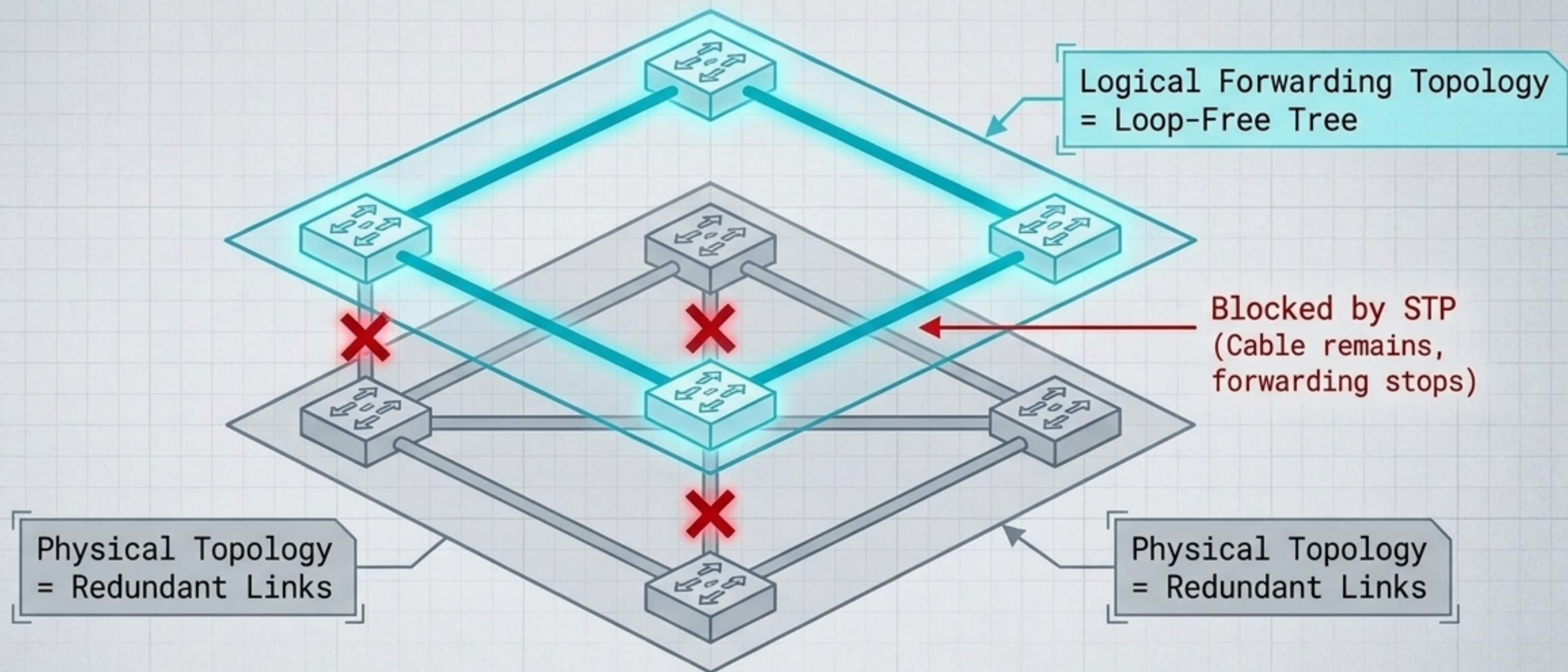


Contradictory topology data triggers MAC address flapping



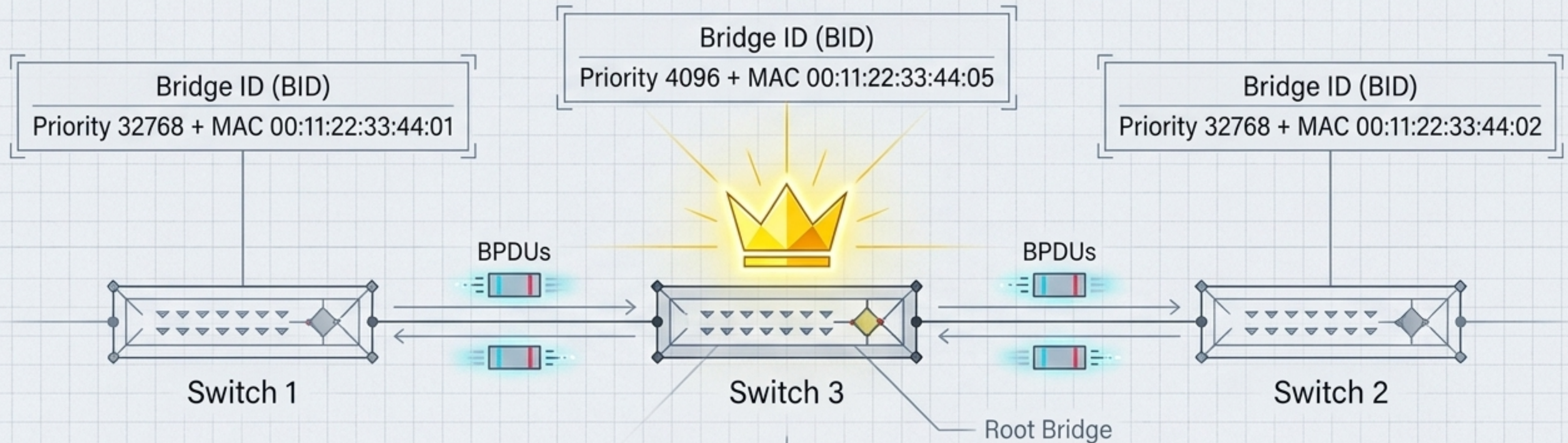
Because frames arrive from multiple directions via redundant loops, the switch continuously overwrites its MAC table, making a stable mapping impossible.

Spanning Tree constructs a logical overlay on physical reality



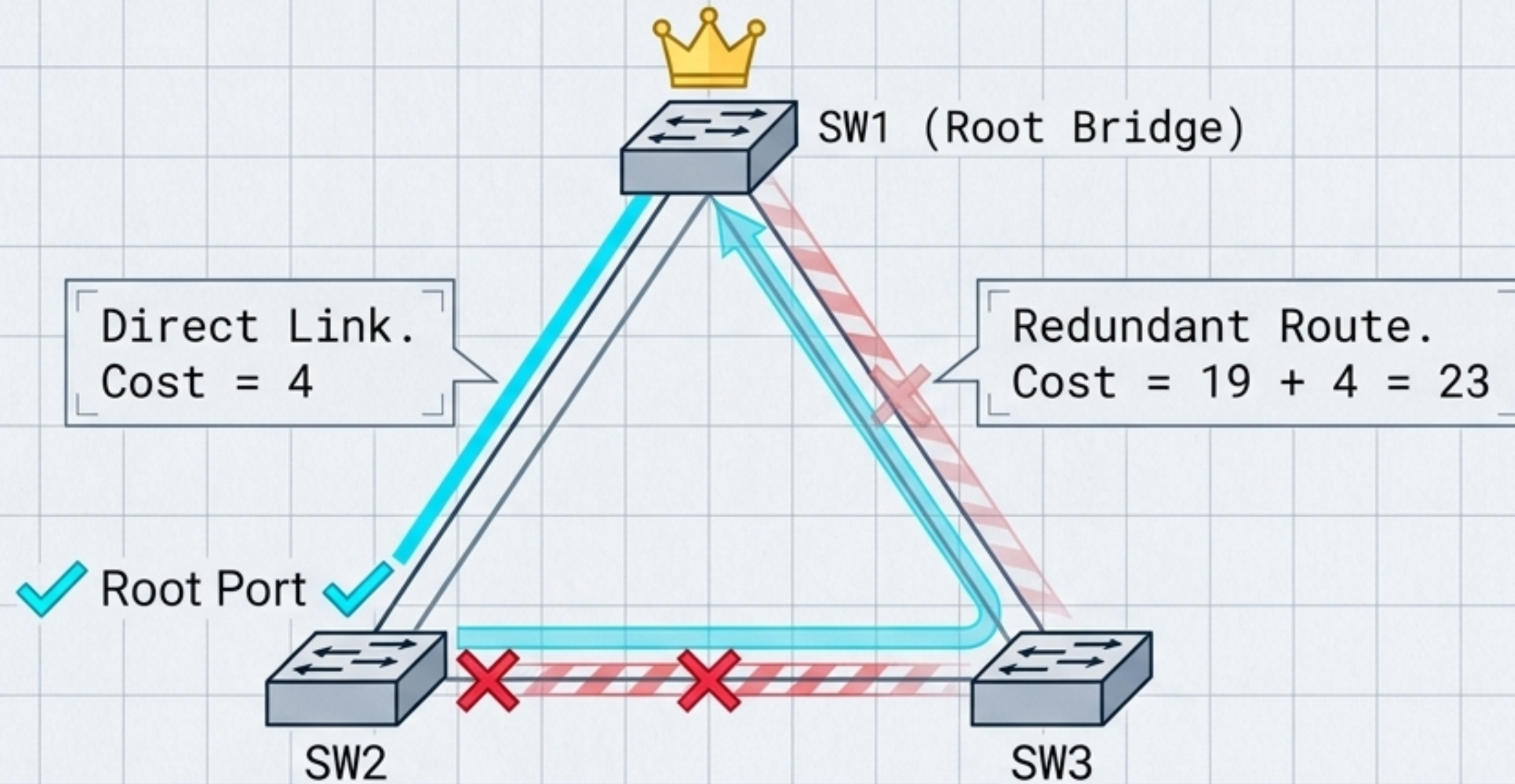
STP solves the paradox by maintaining physical redundancy while mathematically enforcing a single, loop-free active forwarding topology.

The topology anchors around an elected Root Bridge



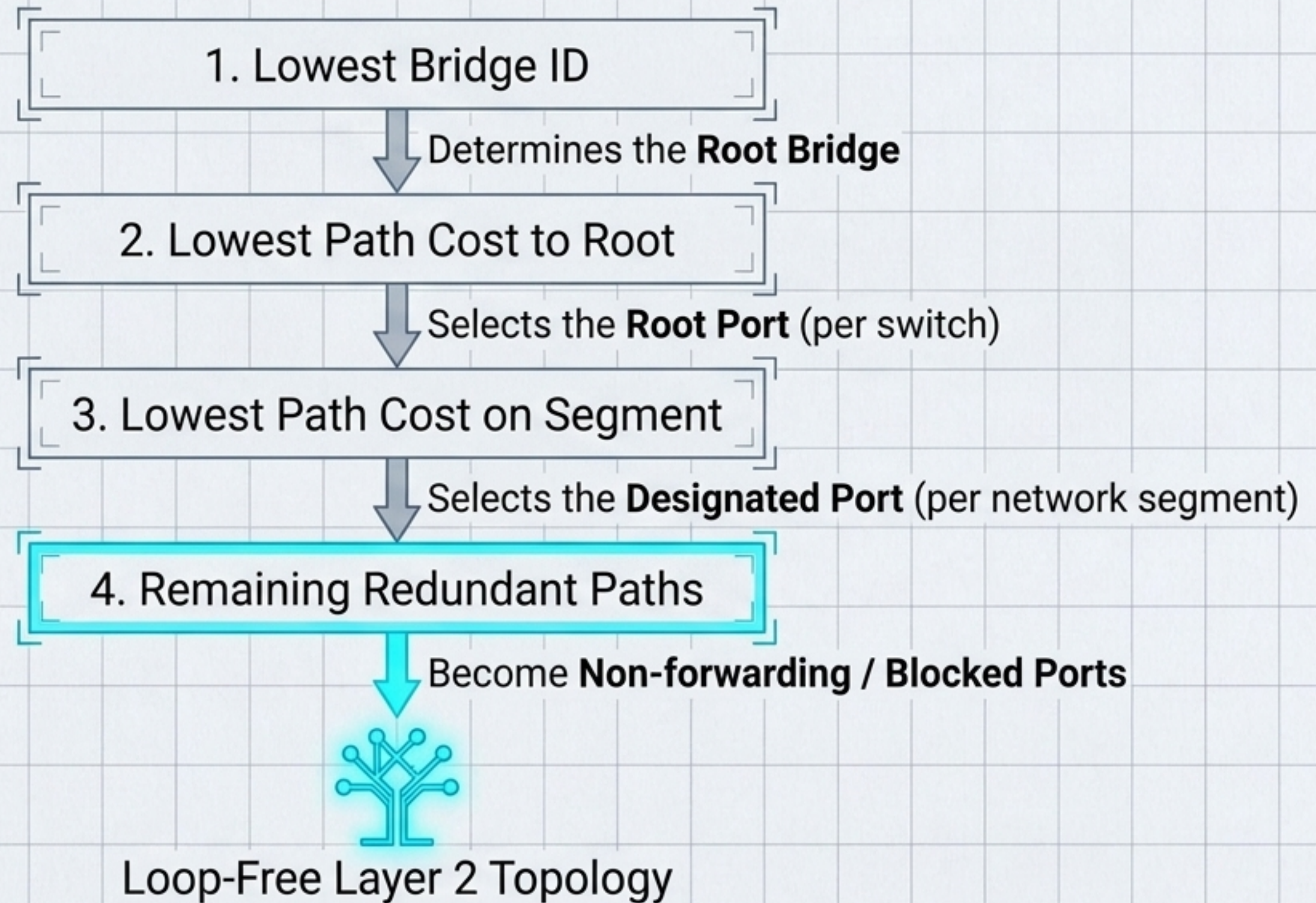
Switches exchange **Bridge Protocol Data Units (BPDUs)**.
The switch with the lowest Bridge ID becomes the Root Bridge—the reference point for all subsequent path calculations.

Switches evaluate cumulative path cost to reach the Root

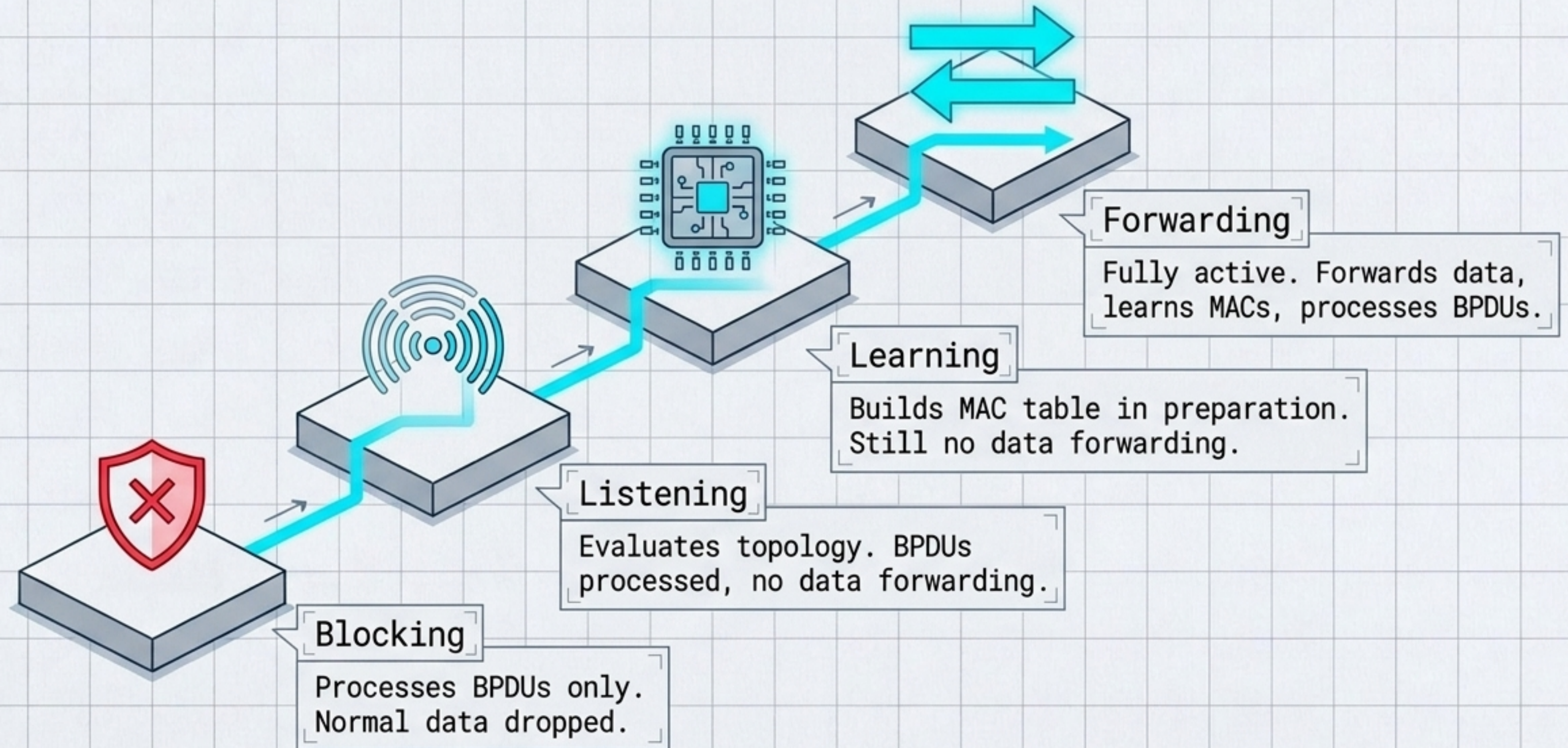


Every non-root switch selects **exactly one Root Port**—the interface with the **lowest total accumulated STP cost** to reach the Root Bridge.

The complete Spanning Tree decision algorithm



Port states escalate carefully to prevent transition loops

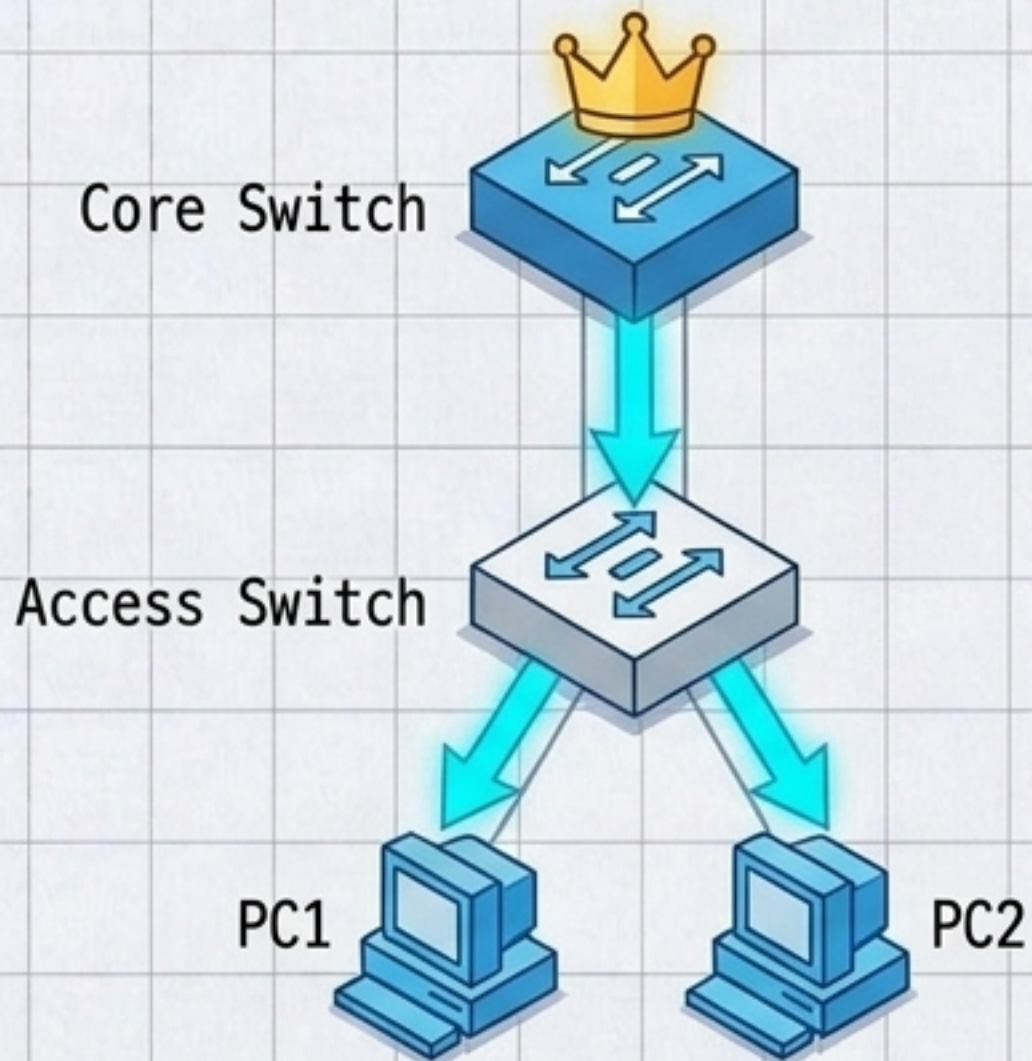


The STP Protocol Evolution Matrix

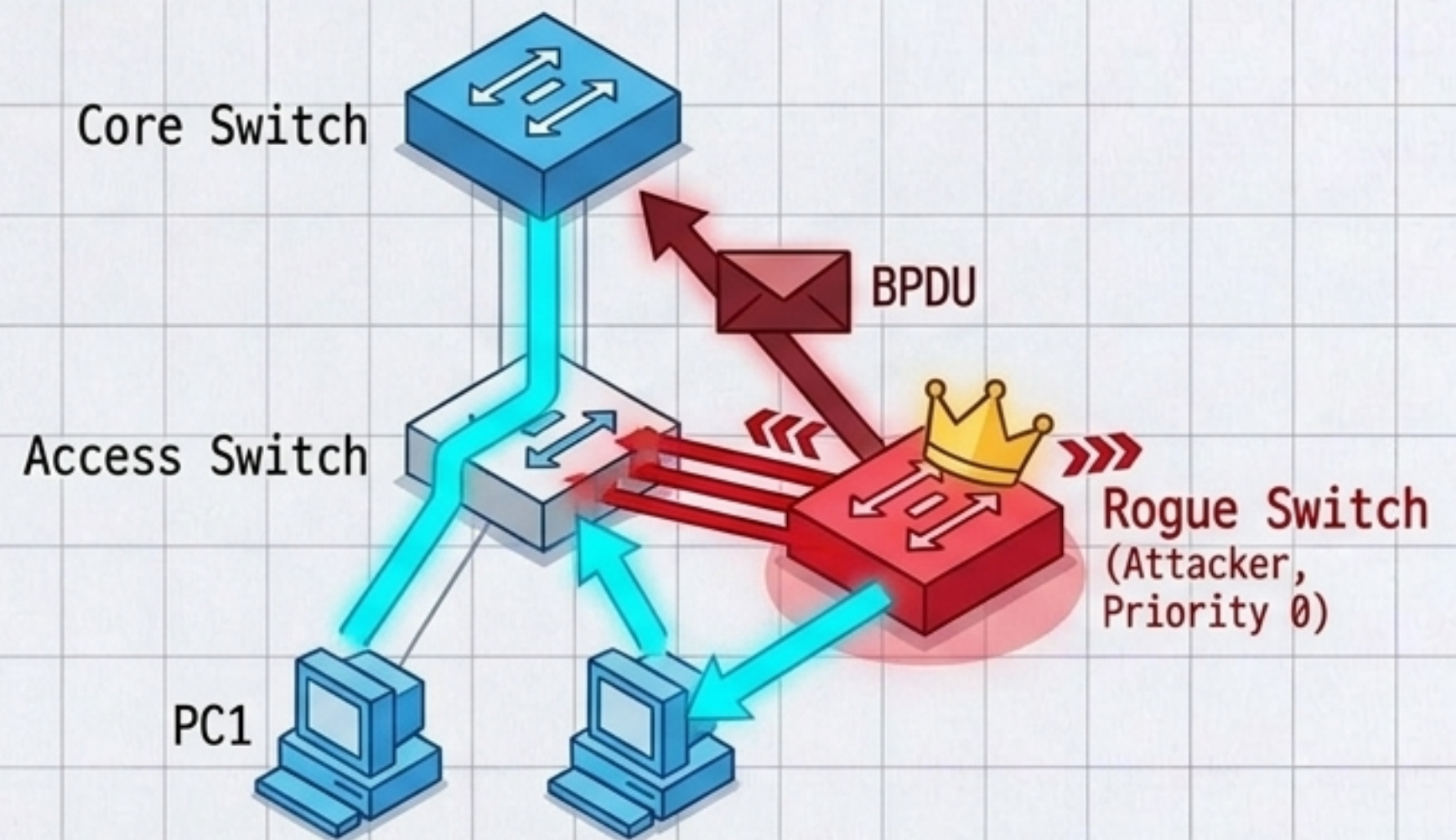
	802.1D (Legacy STP)	802.1w (RSTP)	PVST+ (Cisco Per-VLAN)
Convergence Speed	Slower, relies on timers	<u>Rapid recovery</u>	<u>Varies, generally rapid</u>
Core Mechanism	Single tree for all traffic	<u>Proactive port transitions</u>	<u>Separate spanning-tree instance per VLAN</u>
Link Utilization	✗ Inefficient, redundant links idle	Idle links available faster	<u>High efficiency, different VLANs use different forwarding paths</u>

STP relies on implicit trust, creating a control plane vulnerability

Before: Normal Topology



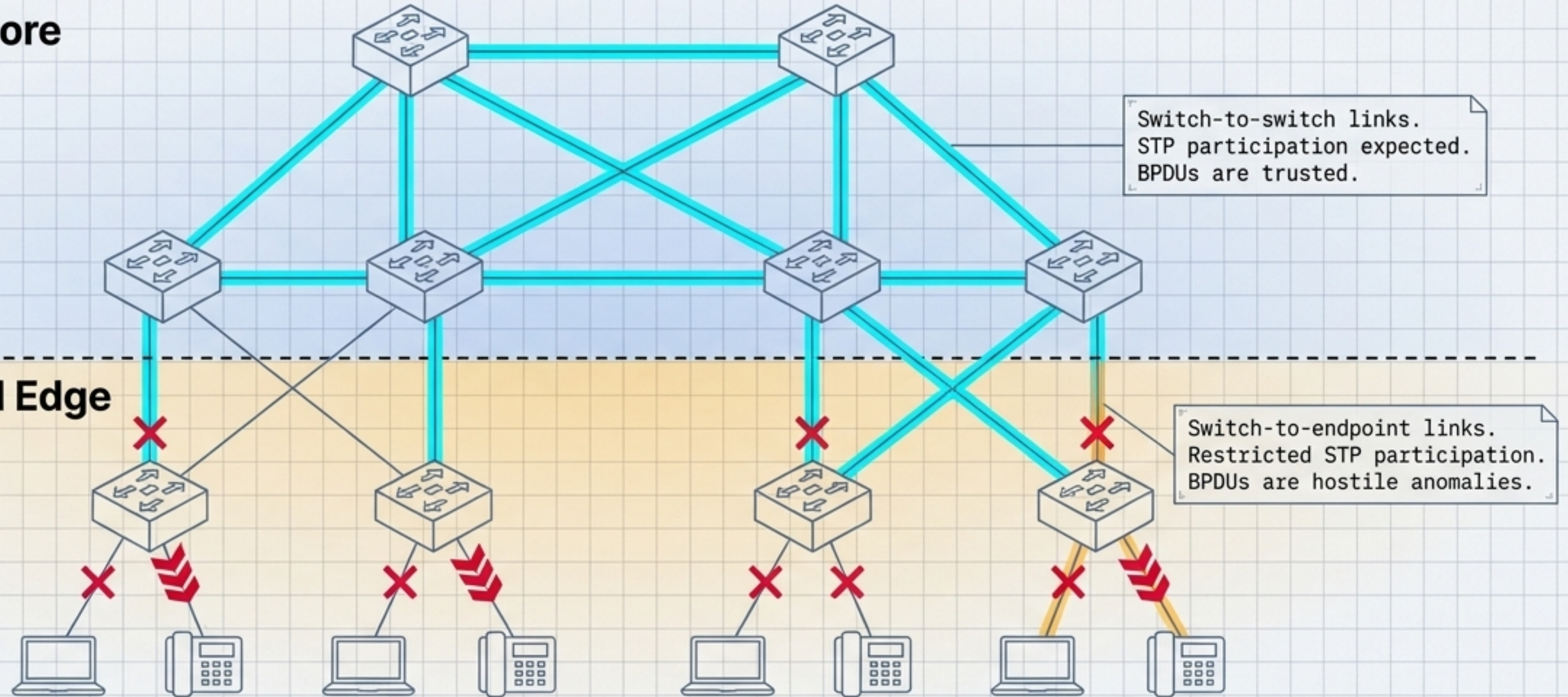
After: The Root Takeover Threat Model



Because STP determines forwarding paths, manipulating the Root election allows attackers to hijack topology, forcing traffic into optimal Man-in-the-Middle (MITM) chokepoints.

Defending the topology requires strict trust boundaries

The Trusted Core (Blue Zone)



Effective Layer 2 defense hinges on **classifying ports**.

Endpoints have **no legitimate reason** to participate in spanning tree calculations.

The Layer 2 Defensive Toolkit Matrix

PortFast

Purpose:

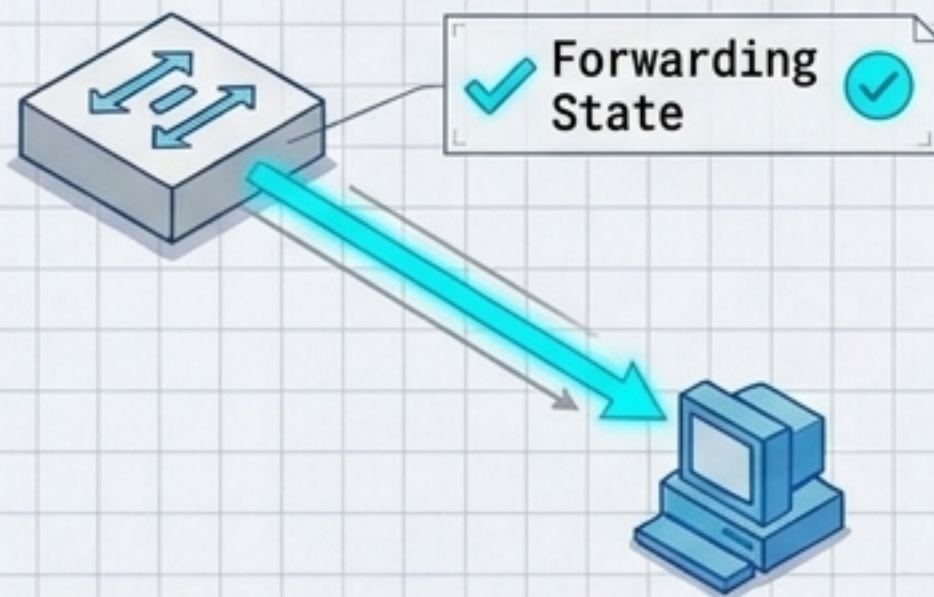
Rapid endpoint activation.

Location:

Edge ports.

Trigger/Action:

Bypasses listening/learning states to immediately enter forwarding.



BPDUGuard

Purpose:

Protect edge from unexpected STP participation.

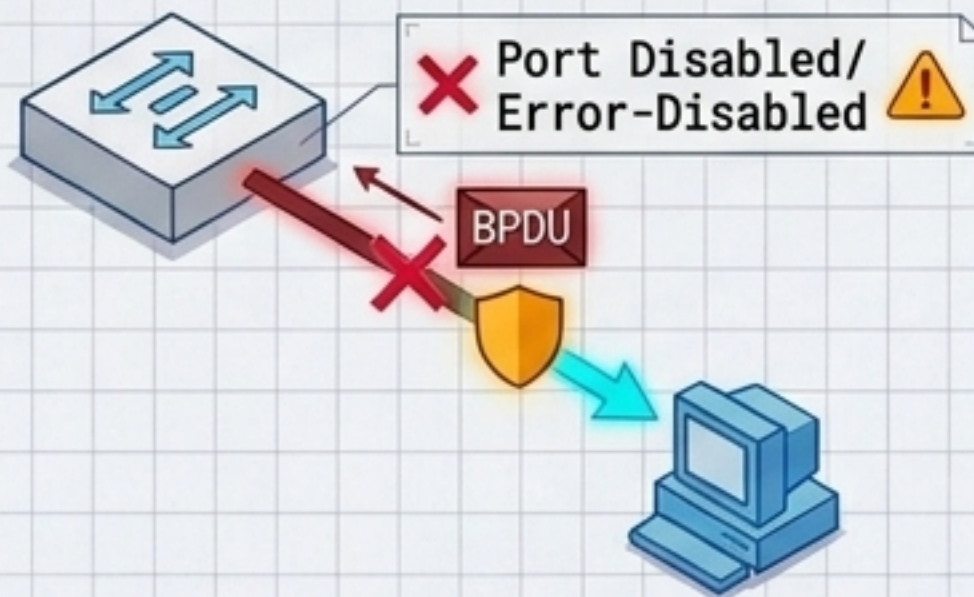
Location:

Edge ports (paired with PortFast).

Trigger/Action:

Trigger: Receiving any BPDU.

Action: Disables/protects port immediately.



Root Guard

Purpose:

Protect the planned Root hierarchy.

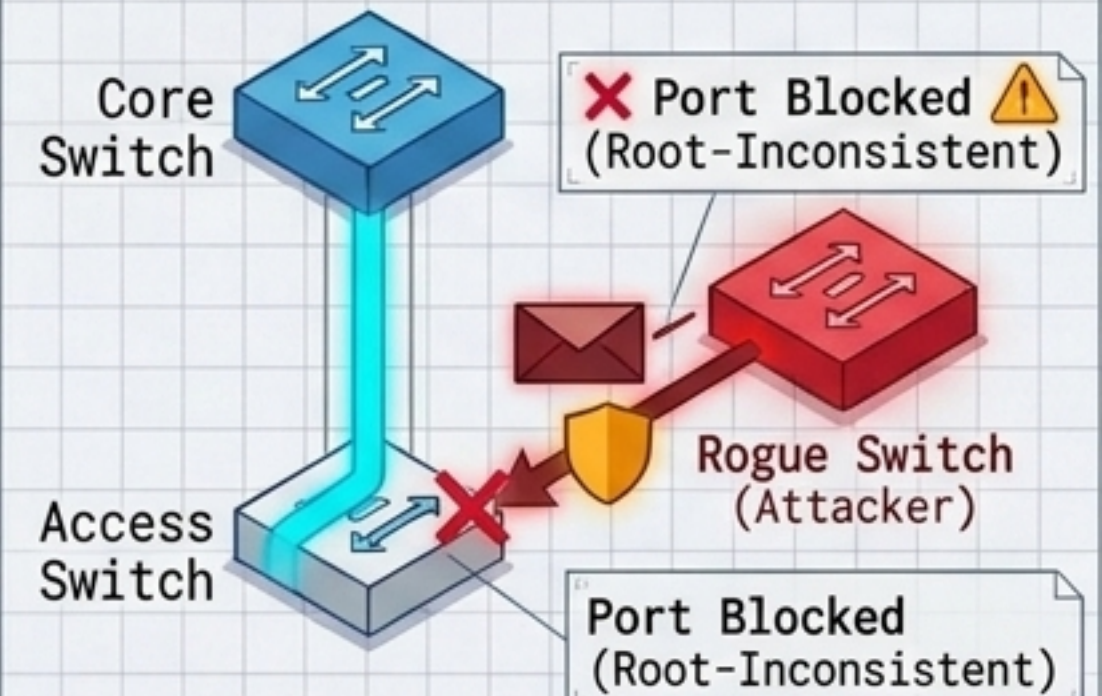
Location:

Downstream switch-to-switch links.

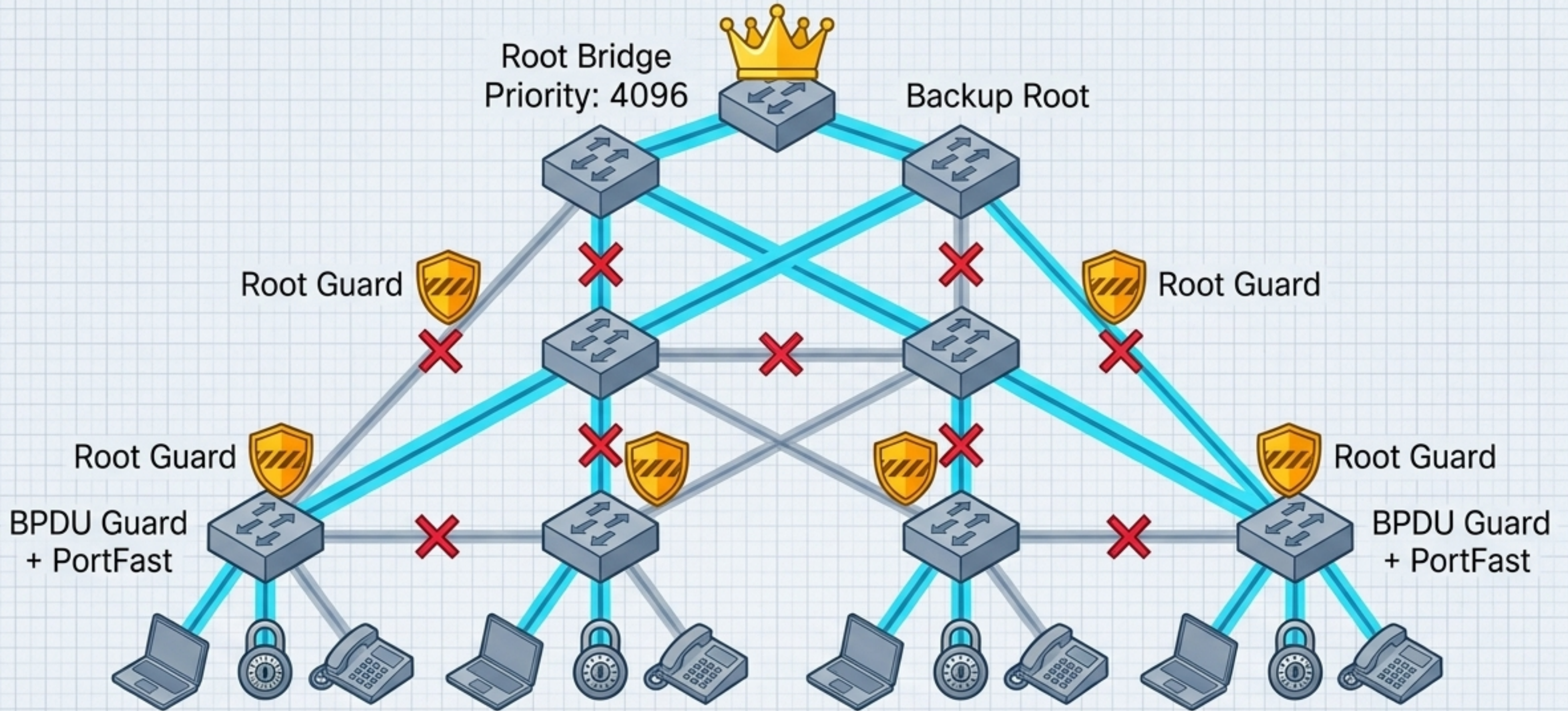
Trigger/Action:

Trigger: Receiving a superior BPDU.

Action: Blocks port to prevent Root takeover.



The Secure Spanning Tree Architecture



Redundancy is essential. Uncontrolled redundancy is catastrophic. A secure Layer 2 network leverages STP to mathematically eliminate loops while aggressively defending the control plane boundaries.